**Yuchen You**
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## Research Interests

Distributed systems, machine learning systems, agentic systems, and computer networking, with a focus on building reliable and robust systems with comprehensive observability and interpretable performance metrics.

## Education

- **University of Maryland** 2026 – 2031 (Expected)
  *Ph.D. in Computer Science* *College Park, MD, US*
  - **Advisor**: *Prof. Zaoxing (Alan) Liu*
- **University of Michigan** 2024 – 2026
  *B.E. in Computer Science* *Ann Arbor, MI, US*
  - **Advisor**: *Prof. Ryan (Peng) Huang, Prof. Xiaonan (Sean) Huang*
  - **Honors**: Cheng-Family Scholarship (Top 2%), Dean's List, Honors List
- **Shanghai Jiao Tong University** 2022 – 2026
  *B.E. in Mechanical Engineering (Graduate with excellent undergrad honor)* *Shanghai, China*
  - **Advisor**: *Prof. Yutong Ban*
  - **Honors**: John Wu & Jane Sun Sunshine Scholarship (Top 6%), SJTU Excellence Scholarship (Top 3%)

## Research Experience

- **OptCC: Straggler-Aware AllReduce** 2026 – Present
  *FrootLab, University of Maryland; Advisor: Prof. Zaoxing (Alan) Liu* *College Park, MD, US*
  - **RDMA testbed and traffic shaping**: Built a reproducible **5-rank RDMA testbed** on CloudLab with **BlueField-2 100 GbE** NICs and automated container deployment. Implemented a **DPDK per-VF rate shaper** on BlueField Arm cores to enforce strict rate limits after identifying leakage in NIC token-bucket caps.
  - **NCCL collective implementation**: Implemented **OptCC AllReduce in NCCL**, including a custom ring kernel, straggler transmit/receive arbitration, staggered launches, and fused primitives for bubble filling. Reimplemented and pipelined the **R2CC baseline** for controlled comparison.
  - **System diagnosis and performance modeling**: Diagnosed ECE-triggered NIC congestion control, spurious retransmissions from ACK timeouts, and shaper queue collisions. Identified a NIC multiplexing assumption in the published lower bound and derived an applicable bound and latency model for the rate-limited testbed.
- **ServeBig: MoE Inference on Consumer GPUs** 2026 – Present
  *FrootLab, University of Maryland; Advisor: Prof. Zaoxing (Alan) Liu* *College Park, MD, US*
  - **MoE serving and evaluation**: Built a reproducible **SWE-bench agent-trace replay benchmark** for Strata and FreeToken at concurrency 1–4. Fixed a **32K-prefill OOM** with bit-exact chunked top-k; added **LRFU expert caching** and **hybrid prefill**, yielding **1.23–1.27× end-to-end speedup** over upstream FreeToken on 28 tasks at concurrency 4 (development branch).
  - **Speculative decoding**: Evaluated **MTP** for offloaded MoE inference and optimized CPU expert kernels with **split-K GEMV** and thread binding; identified CPU/PCIe expert misses and verification overhead as bottlenecks limiting speculation gains.
- **Agentic Distributed System Overload Mitigation** May 2025 – May 2026
  *Order Lab, University of Michigan; Advisor: Prof. Ryan (Peng) Huang* *Ann Arbor, MI, US*
  - **End-to-end system design**: Designed and built a **closed-loop overload mitigation system** for **ZooKeeper** clusters, covering telemetry collection, mitigation actions, workload generation, and evaluation. Integrated **Prometheus**, **HAProxy**, and **tc** to monitor cluster behavior and control traffic.
  - **Agentic diagnosis and control**: Developed an **LLM-based controller** using GPT-4.1 and **MCP tools** to analyze metrics and log summaries, diagnose overload, and invoke auditable mitigation scripts for network throttling, load balancing, JVM heap tuning, and disk I/O control. Implemented log processing to manage context usage.
  - **Workload generation and evaluation**: Built **zkbench** to generate multi-stage workloads (warmup → spike → cooldown), and developed an evaluator to compare static policies with agentic control using **SLO violations** and **goodput** across changing load conditions.

- **Modular Soft Robotic Arm: Software and Control** Sept. 2024 – May 2026
  *HDR Lab, University of Michigan; Advisor: Prof. Xiaonan (Sean) Huang* *Ann Arbor, MI, US*
  - **Complete software stack**: Developed all software for a modular soft robotic arm, spanning **STM32 firmware**, CAN/$I^2C$ communication, and **Orange Pi** planning and control. Implemented isolated real-time tasks, kinematics, and feedback control, achieving **sub-millimeter trajectory tracking accuracy**.

## Publications

- **Origami Inspired Soft Robotic Arm: A Modular Platform for Manipulation** May 2025
  *IEEE ICRA Workshop, Atlanta, USA* *Best Poster Award*
  - **Authors**: Jiyang Wang, **Yuchen You**, Xinqi Zhang, Haobo Fang, Jiaqi Wang, Xiaonan Huang
  - **Award**: Gave a **spotlight talk**; poster received the **Best Poster Award** at the IEEE ICRA 2025 Workshop.

## Selected Projects

- **Search Engine "Crawl Me Maybe" in C++** Jan. 2026 – Apr. 2026
  *EECS 498 (System Design for Search Engine)* *Instructor: Nicole Hamilton, University of Michigan*
  - **Distributed search engine**: Built a **C++ search engine** in a team of four across **20 GCP servers**; crawled 210M websites in two days and built a **1 TB inverted index**, with **200 ms average** and **4 s p99** query latency.
  - **Index storage and retrieval**: Designed a **memory-mapped, multithreaded index** with VByte compression, pooled string storage, skip-list navigation, and positional posting-list cursors; implemented a **zero-copy data path** that made index writing compute-bound.
- **Simulated Distributed System in Golang** Aug. 2025 – Dec. 2025
  *EECS 491 (Introduction to Distributed System)*
  - **Primary–backup KV store**: Implemented a primary–backup key–value store in Go with lexical confinement and request concurrency, designing the protocol to provide linearizable GET/PUT/APPEND operations.
  - **Paxos replication and sharding**: Built a reliable **Paxos-based sharded key–value store** with dynamic reconfiguration and recovery under concurrent shard moves; optimized normal-case latency with accept-phase skipping.
- **Simulated Basic Operating System** Jan. 2025 – Apr. 2025
  *EECS 482 (Introduction to Operating System) Lecture Project*
  - **User-level threading library**: Built a lightweight user-level multicore threading library (context switching, Mesa monitors, synchronization primitives) and designed scheduling to handle concurrent workloads without deadlock.
  - **Pager & virtual memory**: Implemented a minimal pager and MMU layer managing page tables and dirty/reference bits, designing a clock-based eviction policy and copy-on-write support for fork/mmap.

## Skills

- **Programming**: C/C++, Go, Python, Bash
- **Networking**: RDMA; NCCL; tc(config); Mininet; TCP
- **DevOps**: Git; Docker (Compose); ZooKeeper; Prometheus
- **ML**: PyTorch, Triton, CUDA, MCP

## Selected Awards

- Shanghai Jiao Tong University Excellent Graduate Student May 2026
- ICRA Workshop Best Poster Award May 2025
- RoboMaster University Championship: Eastern Region Champion May 2024
- University Physics Competition: Silver Prize Nov. 2023
- RoboMaster University Championship: National Champion Aug. 2023

## Teaching Experience

- **Teaching Assistant (Fall 2026)**: **Teaching Assistant** at University of Maryland, CMSC 424 (Database Design).
- **Teaching Assistant (2024)**: **Teaching Assistant** at Shanghai Jiao Tong University, ENGR 1000J (Introduction to Software Engineering).

## Extra Curriculars

- **Undergraduate Research Assistant (May 2025 – May 2026)**: Undergraduate Research Assistant at University of Michigan CSE Department, MI, USA.
- **ICRA Volunteer (2025)**: **Volunteer** at IEEE International Conference on Robotics and Automation (ICRA), Atlanta, GA, May 2025.